

Visualizing Disruptive Forces

Scenario

This Group Work Project is the culminating assessment for the AAIF 610 Visualizing Disruptive Forces course. It must be submitted at the end of the course and asks your team to bring the full course toolkit to bear on a single real network: the cross-border banking network shown in the Global Banking Network Risk Dashboard (BIS consolidated banking statistics, with the 2008–09 crisis highlighted in the time-series views). Countries are nodes; cross-border lending relationships are weighted, directed edges.

The project is built so that group insight depends on individual work. Every member analyzes the same network, but each owns a different disruptive force and a different analytical lens. No member can complete the group synthesis alone, because the synthesis is produced by overlaying each member's individual analysis onto the shared network and finding where the forces reinforce or offset one another.

Why this Case

The banking network is the one case study that carries an explicit historical crisis in its data, so it exercises causal reasoning (cascade versus common exposure) directly. It is a network by construction, so it exercises all three lenses, and its 2008 dynamics touch systemic, nonlinear, asymmetric, governance, behavioral, and resilience forces at once. Each member can therefore own a distinct force against one shared object.

Working from the Dashboard, not from Hindsight

Because 2008 is one of the most-written-about events in finance, it is easy to lean on published post-mortems instead of reading the tool. This project assesses your ability to read the dashboard. Every empirical claim must cite specific values you read off the Global Banking Network Risk Dashboard — named countries, lending figures, connectivity scores, time-series points — rather than outside commentary. Outside sources may frame context but may not substitute for evidence drawn from the dashboard itself.

Tasks - Groups of Three Members

Step 1: Individual Foundation

Each group member takes one of the three roles listed below and develops a 2-3 pages document individually.

Each of the deliverables below must ground its empirical claims in specific values read off the dashboard (i.e., named countries, lending figures, connectivity scores, or time-series points), not in outside commentary about 2008.

Member A – The Network Cartographer

Force focus: Systemic risk (Module 1) and Resilience (Module 7).

Primary lens: Network thinking

Draft 2-3 pages document with the following:

- **A1:** Define systemic risk and resilience in your own words. From the dashboard, build a node table identifying the top 5 hub countries by lending connections, 3 peripheral nodes, and any bridges between regional clusters. Name hubs, paths, clusters, bridges, and robustness explicitly.
- **A2:** Assess the connection–cascade trade-off: what efficiency did connectivity buy, and what cascade risk did it create? Propose one “circuit breaker” placement and justify it in network terms. Classify the network's resilience form as robustness, adaptive capacity, or transformability.

Member B – The Causal Analyst

Force focus: Nonlinear risk (Module 2) and Asymmetric risk (Module 3).

Primary lens: Causal inference

Draft 2-3 pages document with the following:

- **B1:** Define nonlinear and asymmetric risk in your own words and name their visual signatures (the hockey-stick curve and the fat-tailed or skewed distribution). From the 2008–09 time series, identify one candidate threshold and one fat-tail feature where a few nodes dominate exposure.
 - **B2:** Apply the “but for” test to one 2008 cascade path. Distinguish a genuine cascade ($A \rightarrow B \rightarrow C$) from a common exposure (A and B both depend on C). Name one confounder and one selection- or survivorship-bias risk in the data.
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Member C — The Strategic-Behavior Analyst

Force focus: Governance and policy risk (Module 4) and Behavioral risk (Module 6), with a light technological-risk touchpoint (Module 5).

Primary lens: Basic game theory

Draft 2-3 pages document with the following:

- **C1:** Define governance/policy risk and behavioral risk in your own words and name the strategic patterns (coordination failure, prisoner's dilemma, dominance). From the dashboard, identify one situation where an actor's best move depends on what others do, such as withdrawing cross-border funding in a panic.
- **C2:** Model one 2008 episode as a coordination or prisoner's-dilemma problem: who are the players, what are the incentives, and what is the bad equilibrium (bank-run or herding logic)? Identify one governance intervention and one way actors might game it.

Step 2: Integration

As a group, complete the following (up to:

- **2A — Overlay map:** Combine Member A's network map, member B's causal cascade paths, and member C's strategic-behavior nodes onto one annotated diagram of the shared network (as a full-page figure). Mark the triple-points where a hub sits on a cascade path and is also a strategic actor; these are the team's key findings.
- **2B — Cross-force interaction table:** Build a grid (table) of the forces your team covered, and for each pair state whether they reinforce or offset one another on this network, with one sentence of evidence per cell.
- **2C — Cross-domain confirmation:** Choose one secondary dashboard (Supply Chain, Wind Power, COVID Evidence Explorer, and so on) and show in one paragraph and one figure that the same structural pattern recurs in a different domain.

Step 3: Group Evaluation and Recommendation

As a group, complete the following:

- **3A – Trade-off resolution:** Identify the single most important cross-lens trade-off the team found, for example a resilience circuit-breaker that worsens a coordination problem. Argue and justify a resolution.
- **3B – Integrated disruptive-risk assessment:** Assess the network as if it were an organization: which forces dominate, which lens is most decision-relevant, and what one intervention the team recommends.

Step 4: Non-technical Briefing

Write a one-page briefing to a non-technical decision-maker (for example, a finance-ministry official). Explain the biggest disruptive risk in the shared network, recommend one action within their authority, and include one visual that communicates the key lesson at a glance. Plain language, no jargon, credible.

Page allowances for Steps 2, 3, and 4: up to roughly 18 pages plus figures. Groups exceeding this length will be subject to penalties.

Step 5: Individual Reflection

Each member writes half a page: which lens you owned, what overlaying your work with your teammates' revealed that your solo analysis missed, and one thing you would re-examine. This enforces individual accountability and makes explicit that the group insight depended on each individual contribution.